Q. No.	Questions
UNIT 1	**SECTION 1**
1	A pharmaceutical company evaluates the effect of a new anticancer drug on cultured cells. DNA sequencing before and after treatment shows no genomic alterations, but RNA sequencing and proteomic analysis indicate significant molecular changes. Interpret the relationship between the genome, transcriptome, and proteome, and discuss why drug treatment alters RNA and protein profiles without changing the genome. (5 Marks)
2	A biotechnology laboratory compares liver cells and skeletal muscle cells obtained from the same healthy individual. Whole-genome sequencing confirms that both cell types possess identical DNA sequences, whereas transcriptome and proteomic analyses reveal distinct RNA and protein expression profiles. Compare the genome, transcriptome, and proteome of the two cell types, and explain why the transcriptome and proteome differ despite having an identical genome. (5 Marks)
3	A biotechnology laboratory transforms E. coli with a recombinant plasmid carrying an ampicillin-resistance gene and a gene encoding human insulin. After recovery, the transformed cells are plated on LB agar containing ampicillin. The next day, only four colonies are observed, while the negative control plate shows no growth. The transformation protocol, competent cells and antibiotic concentration were reported to be correct. Interpret the experimental outcome based on the observations. (5 Marks)
4	(i) A scientist plans to express a human gene in E. coli. The genomic DNA contains several introns, but the cloned DNA produces the correct protein in bacteria. Infer the DNA source used for cloning. (3 Marks)
	(ii) A transformed E. coli culture is allowed to grow overnight. Analysis shows that multiple copies of the inserted gene are present in each bacterial cell. Interpret the observation. (2 Marks)
UNIT 1	**SECTION 2**
5	A laboratory receives DNA samples from patients with a rare inherited disorder. Before starting wet-lab experiments, researchers identify candidate genes, retrieve chromosomal locations and compare reference DNA sequences using existing genomic resources. Interpret how previous large-scale genome research supports this workflow. (5 Marks)
6	A genomics laboratory follows the workflow below: Patient DNA -> Variant Detection -> Chromosomal Localization -> Gene Annotation -> Candidate Gene Identification Researchers complete the entire analysis within a short period without performing additional genome sequencing. Infer the genomic resources that support each stage of this workflow. (5 Marks)
7	A newly discovered protein contains a stable -helical region and a flexible disordered region. Experimental studies show that the protein interacts with multiple regulatory partners inside the cell. Interpret the structural organization of the protein. (5 Marks)
8	(i) A research laboratory studies the stability of a purified enzyme by exposing it to a urea-containing solution. The enzyme loses its native structure during treatment but regains its original activity after the urea is removed. Interpret the observations. (3 Marks)
	(ii) A protein sample is heated during purification. The protein loses its biological activity but its peptide bonds remain intact. Infer the molecular change responsible for this observations.

	(2 Marks)
UNIT 2	**SECTION 1**
9	A diagnostic centre introduces next-generation sequencing for analysing samples from patients with inherited genetic disorders. The generated sequence data is interpreted to identify disease-associated genetic changes. Apply the principles of NGS in analysing sequence data for genetic disease identification. (5 Marks)
10	A plant breeding group identifies a yield-associated trait in a crop genome containing recombination variation and incomplete sequence information. The researchers combine different mapping approaches to locate the candidate gene. Select an integrated mapping strategy for identifying a yield-related gene in the crop genome and justify your answer. (5 Marks)
11	A crop improvement programme compares salt-tolerant and salt-sensitive varieties and detects numerous single-base variations across the genome. The researchers use these variations to identify genomic regions associated with the trait. Interpret how SNPs can be applied to investigate complex traits in crop improvement. (5 Marks)
12	(i) A DNA analysis compares the same chromosomal region in two individuals and identifies a difference at only one nucleotide position. The observed variation is further analysed for genetic comparison. Interpret the genetic variation identified in this analysis. (3 Marks)
	(ii) Researchers analyse DNA markers in family members to determine the inheritance pattern of a genetic trait. SNP markers are selected throughout the chromosome during the analysis. Identify the application of SNPs in this investigation. (2 Marks)
UNIT 2	**SECTION 2**
13	A laboratory prepares a cDNA library from developing flower tissue to identify genes involved in flower development. The obtained Expressed Sequence Tags (ESTs) sequences are compared with a reference genome. Apply EST data to identify expressed genes in the developing flower. (5 Marks)
14	A family affected by the same inherited disorder undergoes genetic analysis, and different genetic variants are identified among affected individuals. Researchers combine clinical observations and molecular evidence to establish the relationship between genes and the disorder. Apply the evidence used to establish gene-disease association for the identified disorder. (5 Marks)
15	A genomics laboratory identifies five candidate genes after sequencing patients affected by a rare disorder. Further investigation shows: Gene A: Highly expressed but found equally in healthy individuals. Gene B: Rare variant but unrelated biological function. Gene C: Variant segregates with the disease in all affected family members and functional analysis shows reduced protein activity. Gene D: Common population variant. Gene E: No detectable expression in the affected tissue. Determine the candidate gene most strongly associated with the disorder and support your conclusion using the available evidence. (5 Marks)
16	(i) A patient is scheduled for a genetic test but has not received sufficient information about the possible outcomes of the investigation. The healthcare team reviews the testing procedure before sample collection. Apply the appropriate practices before conducting the genetic test. (3 Marks)
	(ii) A person's genomic information is shared without permission, resulting in discrimination during employment. The incident is reviewed according to ethical principles in genome research. Identify the ethical issues demonstrated in this situation.

	(2 Marks)
UNIT 3	**SECTION 1**
17	A pharmaceutical company evaluates a newly developed anticancer drug using tumour samples collected before and after treatment. Genome sequencing shows no significant DNA changes, but proteomic analysis reveals that several signalling proteins become downregulated after therapy, while others remain unchanged. Assess how proteomic findings contribute to evaluating the effectiveness of the anticancer drug. (5 Marks)
18	A biotechnology company is purifying a therapeutic protein from a cell lysate. The target protein specifically binds to a particular ligand, while most contaminating proteins do not. Select the most suitable chromatography technique and justify the reason it is preferred over the other two techniques. (5 Marks)
19	A research laboratory compares protein extracts from healthy and diseased tissues using two-dimensional gel electrophoresis. Several additional protein spots appear only in the diseased sample. Interpret the observation and justify how electrophoretic analysis supports proteome investigation. (5 Marks)
20	(i) A biotechnology laboratory compares protein samples collected before and after drug treatment using SDS-PAGE. Additional protein bands are detected after treatment. Interpret the observations in relation to protein expression. (3 Marks)
	(ii) During repeated analysis, one protein band becomes darker after treatment while its position on the gel remains unchanged. Interpret the observation. (2 Marks)